Challenge: Signals from the Void

Narrative:

`INTERCEPTED TRANSMISSION - ORION DEEP SPACE RELAY - SIGNAL ORIGIN: UNKNOWN`

At 0300 hours, contact with the Orion Deep Space Relay was lost. The last known operator - callsign POLARIS - went dark without warning. What they left behind was this: a stream of anomalous telemetry data, still trickling in through the relay's FTP archive. Someone, or something, is hiding a message inside it. Your team has been tasked with intercepting and decoding it.

Objective

The goal of this challenge is to intercept and decode the covert message being transmitted through the **Orion Deep Space Relay**. The relay's FTP archive encodes a hidden message by creating files and directories and setting their permissions to represent binary data. The message is restricted to characters in the basic ASCII character set (values 0–127).

Whether the covert message is encoded using all 10 permission bits of the files and directories on the relay (or just in files using the right-most 7 bits, or both) must be determined. Moreover, where these entries are located in the relay's archive must also be determined. Not everything you see is signal — some of it is noise. Explore. Tinker. The cosmos rewards the patient.

Your decoded message must match the current message being transmitted since it changes every minute. When you think that you have successfully completed the challenge, tag the **Oberon/prof** team on your private Discord channel. This will be used to determine the order in which the teams completed the challenge.

As a challenge deliverable, upload a ZIP file containing the following items to Canvas:

(1) A pdf document that provides a thorough summary of your team's completion of the challenge. The document must include:
   - the decoded message;
   - the time at which you obtained and decoded the message;
   - a thorough description of the steps you took to obtain the message; and
   - a description of **each** team member's contributions. Please note that each team member should contribute their own paragraph to this section.

(2) Any modified code (e.g., your FTP program, Vigenere program, Binary program); and

(3) Any other scripts or programs that you created to assist you (if applicable).

You will be graded on how well you:

(1) Determine the FTP login information (5%);
(2) Determine the FTP login credentials (10%);
(3) Obtain and decode the covert message (matching the current time) (40%);
(4) Document your process (40%); and
(5) Submit the required ZIP file (5%).

Good luck!